12TH BIOLOGY · CHAPTER 14 · SOLVED EXERCISE

# Human Urinary System

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

## SECTION 1: MULTIPLE CHOICE QUESTIONS

**1. Which one of these is not a part of single nephron?**

(a) Distal convoluted tubule
(b) Collecting duct
(c) Bowman’s capsule
(d) Loop of Henle

**Answer: (b) Collecting duct** — The distal convoluted tubules of many nephrons drain into one collecting duct, so it is shared and is not a part of a single nephron.

**2. In a healthy person, glucose is present in blood but not in urine. This is because glucose molecules are;**

(a) Reabsorbed from the proximal convoluted tubule to blood
(b) Oxidised in kidneys
(c) Stored in kidneys
(d) Too large to enter Bowman's capsule

**Answer: (a) Reabsorbed from the proximal convoluted tubule to blood** — Glucose is filtered into the Bowman’s capsule, but it is returned from the proximal tubule to the blood by active transport.

**3. The sizes of molecules present in Bowman's capsule are smaller than the sizes of molecules present in;**

(a) Afferent renal arteriole
(b) Collecting duct
(c) Loop of Henle
(d) Proximal tubule

**Answer: (a) Afferent renal arteriole** — Blood in the afferent arteriole also contains large plasma proteins and blood cells. These cannot pass into the Bowman’s capsule; only small molecules are filtered.

**4. If a drug reduces mitochondrial activity in nephrons, which chemical will be present in increased amounts in the urine?**

(a) Ammonia
(b) Glucose
(c) Uric acid
(d) Urea

**Answer: (b) Glucose** — Glucose is reabsorbed by active transport, which needs energy (ATP) from mitochondria. With less ATP, glucose is not fully reabsorbed and appears in urine.

**5. The water content of blood is regulated by ADH. In which part of the nephron this regulation occurs?**

(a) Bowman's capsule
(b) Proximal convoluted tubule
(c) Loop of Henle
(d) Collecting duct

**Answer: (d) Collecting duct** — ADH acts on the walls of the collecting ducts and makes them more permeable to water.

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**6. In kidneys, glucose is reabsorbed into blood. Where does this reabsorption occur?**

(a) Bowman's capsule
(b) Glomerulus
(c) Proximal convoluted tubule
(d) Collecting duct

**Answer: (c) Proximal convoluted tubule** — Glucose, amino acids and salts are returned to the blood from the proximal convoluted tubule by active transport.

**7. In normal conditions, which one is completely reabsorbed from renal tubule into blood?**

(a) Urea
(b) Water
(c) Salts
(d) Glucose

**Answer: (d) Glucose** — Glucose is a vital nutrient, so all of it is reabsorbed. Water and salts are reabsorbed only partly, and urea is excreted.

**8. If the loops of Henle are absent in all nephrons, what would happen?**

(a) No urine formation
(b) Formation of urine with normal concentration
(c) Formation of concentrated urine
(d) Formation of dilute urine

**Answer: (d) Formation of dilute urine** — The loop of Henle creates the high salt concentration in the surrounding fluid (counter-current mechanism). Without it, water cannot be reabsorbed from the collecting duct, so the urine is dilute.

**9. In peritoneal dialysis, the lining of which part of the body acts as a filter?**

(a) Stomach
(b) Intestine
(c) Lungs
(d) Abdomen

**Answer: (d) Abdomen** — Peritoneal dialysis uses the peritoneal membrane that lines the abdominal cavity.

**10. During which of these treatments, the patients are prescribed immunosuppressant drugs?**

(a) Lithotripsy
(b) Haemodialysis
(c) Kidney transplant
(d) Peritoneal dialysis

**Answer: (c) Kidney transplant** — After a kidney transplant, immunosuppressant medications are given to prevent the immune system from rejecting the new kidney.

## SECTION 2: SHORT QUESTIONS

**1. Justify the functioning of kidneys as both excretion and osmoregulation.**

**Excretion:** Kidneys filter metabolic wastes (such as urea) and extra water from the blood and excrete them in the form of urine.

**Osmoregulation:** While doing the excretory function, kidneys also control the water and salt concentrations of body fluids. When body fluids are hypotonic (much water),

they form dilute urine. When body fluids are hypertonic and there is dehydration, they form concentrated urine. So the kidneys are both excretory and osmoregulatory organs.

### 2. List the organs of the urinary system.

The human urinary system consists of:

- **Two kidneys** — filter the blood and form urine.
- **Two ureters** — transport urine from the kidneys to the urinary bladder.
- **One urinary bladder** — serves as the urine reservoir.
- **One urethra** — carries urine from the urinary bladder to the outside of the body.

### 3. Define glomerular filtration.

Glomerular filtration (ultrafiltration) is the first step of urine formation. It is a physical process in which the high blood pressure in the glomerulus forces water, urea, glucose, amino acids, vitamins and various salts from the blood into the Bowman's capsule. About one-fifth (20%) of the fluid portion of the blood is filtered and is called glomerular filtrate. Plasma proteins and blood cells are too large to pass, so they remain in the blood.

### 4. Compare the function of glomerular capillaries and peritubular capillaries.

| Feature | Glomerular capillaries | Peritubular capillaries |
|---|---|---|
| Location | Within the Bowman's capsule | Around the renal tubules |
| Formed from | Afferent arteriole | Efferent arteriole |
| Pressure | High blood pressure | Low hydrostatic pressure |
| Function | Filtration: water and small solutes are forced out of the blood into the nephron | Reabsorption and secretion: nutrients (glucose, ions) and water are pulled back into the blood; specific toxins are secreted into the tubule |

### 5. What are the three main processes involved with urine formation?

1. **Glomerular filtration:** High pressure in the glomerulus forces water and small solutes from the blood into the Bowman's capsule.
2. **Selective reabsorption:** Useful materials (water, glucose, amino acids, salts) are transported from the renal tubule back into the peritubular capillaries.
3. **Tubular secretion:** The cells of the distal convoluted tubule take hydrogen ions, ammonium, potassium ions, toxins and drug residues from the blood and secrete them into the filtrate.

### 6. What are the causes of kidney failure?

Kidney (renal) failure is a condition in which the kidneys lose their ability to filter wastes from the blood and to regulate the fluid and salt balance of the body.

- **Chronic kidney failure** is often caused by underlying conditions like diabetes, high blood pressure or kidney disease (chronic infection, inflammation of glomeruli, etc.).

- **Acute kidney failure** may be due to kidney infection, blockage in the urinary tract (due to kidney stones, bladder cancer or enlarged prostate), certain medications such as NSAIDs, and blood clots in the renal arteries or arterioles.

### 7. Why do blood and dialysate flow in opposite direction?

In the dialyzer, the blood and the dialysate flow in opposite directions on the two sides of the semipermeable membrane. This is a counter-current flow. Due to it, the blood always meets a dialysate that has a lower concentration of wastes than the blood. So a concentration gradient is maintained along the whole length of the membrane, and the wastes keep diffusing from the blood into the dialysate. In this way, the maximum amount of wastes is removed from the blood.

*The textbook describes the dialyzer but does not give this reason. The answer is based on the standard principle of counter-current exchange.*

### 8. Differentiate between:

#### • Afferent and efferent arterioles

| Afferent arteriole | Efferent arteriole |
|---|---|
| It arises from the branches of the renal artery. | It arises from the glomerular capillaries. |
| It carries blood into the glomerulus. | It carries blood away from the glomerulus. |
| It is wider. | It is narrower. This keeps the blood in the glomerulus under high pressure. |
| It leads into the glomerular capillaries. | It branches into the peritubular capillaries (and vasa recta in juxtamedullary nephrons). |

#### • Chronic and acute renal failure

| Chronic renal failure | Acute renal failure |
|---|---|
| It develops over a longer period of time. | It occurs suddenly. It is also called kidney injury. |
| It is irreversible. | It often resolves with treatment. |
| It is often caused by diabetes, high blood pressure or kidney disease (chronic infection, inflammation of glomeruli). | It may be due to kidney infection, blockage in the urinary tract, certain medications such as NSAIDs, and blood clots in renal arteries or arterioles. |

### • Haemodialysis and peritoneal dialysis

| Haemodialysis | Peritoneal dialysis |
|---|---|
| Blood is circulated outside the body through an equipment called a dialyzer. | Blood is cleaned inside the body, in the abdominal cavity. |
| The filter is a semipermeable membrane made of cellulose. | The filter is the peritoneal membrane that lines the abdominal cavity. |
| Dialysate is present in the dialyzer. | 1.5 to 2 litres of dialysate is infused into the abdomen through a catheter and left for 4 to 6 hours. |
| Catheters are inserted into an artery and a vein, usually in the arm. | A catheter is inserted into the abdomen. |
| Hypotension is a common complication. | The process is repeated several times a day. |

### • Renal cortex and renal medulla

| Renal cortex | Renal medulla |
|---|---|
| It is the outer portion of the kidney. | It is the inner portion of the kidney. |
| It makes up about one-third of the kidney's mass. | It makes up two-thirds of the kidney. |
| Cortical nephrons (about 85%) are located primarily in it. | It consists of cone-shaped renal pyramids. The long loops of Henle of juxtamedullary nephrons extend deep into it. |
| Peritubular capillaries are present around the convoluted tubules in the cortex. | Vasa recta run parallel to the loops of Henle in the medulla. |

## SECTION 3: LONG QUESTIONS

### 1. Describe the structure of kidney and relate it with its function.

#### External structure

A kidney is a reddish-brown, bean-shaped organ. Each kidney has an adrenal gland on its top. The kidney and its adrenal gland are surrounded by a fibrous **renal capsule**. Each kidney has a convex and a concave side. A depressed area on the concave side is the **renal hilum** (hilus). At this point, the renal artery enters the kidney, and the renal vein and ureter leave it.

#### Internal structure

- **Renal cortex:** It is the outer portion. It makes up about one-third of the kidney's mass.
- **Renal medulla:** It is the inner portion. It makes up two-thirds of the kidney. It consists of a number of cone-shaped **renal pyramids**.

- **Renal pelvis:** It is a funnel-shaped structure in the centre of the kidney. Urine is collected in the renal pelvis and leaves the kidney through the ureter.
- **Nephrons:** Each kidney contains about one million microscopic tubular structures called nephrons. These are the functional units of the kidney.

### Relation of structure with function

- The renal artery brings blood to the kidney under pressure. The kidneys receive about 20–25% of the total cardiac output, so a large volume of blood is filtered.
- About one million nephrons in each kidney provide a very large surface for filtration, reabsorption and secretion.
- The cortex contains the renal corpuscles and convoluted tubules, where filtration and most of the reabsorption take place.
- The medulla contains the loops of Henle and collecting ducts, which make the surrounding fluid salty. This helps in the reabsorption of water and in the formation of concentrated urine.
- The funnel-shaped renal pelvis collects the urine and passes it into the ureter.

## 2. Explain the detailed structure of nephron.

Each kidney contains about one million microscopic tubular structures called **nephrons**. These are the functional units of the kidney. A nephron consists of two main portions: a renal corpuscle and a renal tubule.

### 1. Renal corpuscle

It consists of two associated structures:

- **Glomerulus:** It is a capillary network derived from an afferent renal arteriole.
- **Bowman's capsule:** It is a cup-shaped, double-walled sac that surrounds the glomerulus.

### 2. Renal tubule

It is a long tubule that leads away from the Bowman's capsule. It consists of three connected portions:

- **Proximal convoluted tubule:** It is the beginning portion and is highly coiled. It dips into the medulla.
- **Loop of Henle:** It is a U-shaped hairpin turn with two limbs. The descending limb continues toward the medulla. The ascending limb returns toward the cortex and reaches parallel to the renal corpuscle.
- **Distal convoluted tubule:** The ascending limb opens into it. It is highly coiled again.

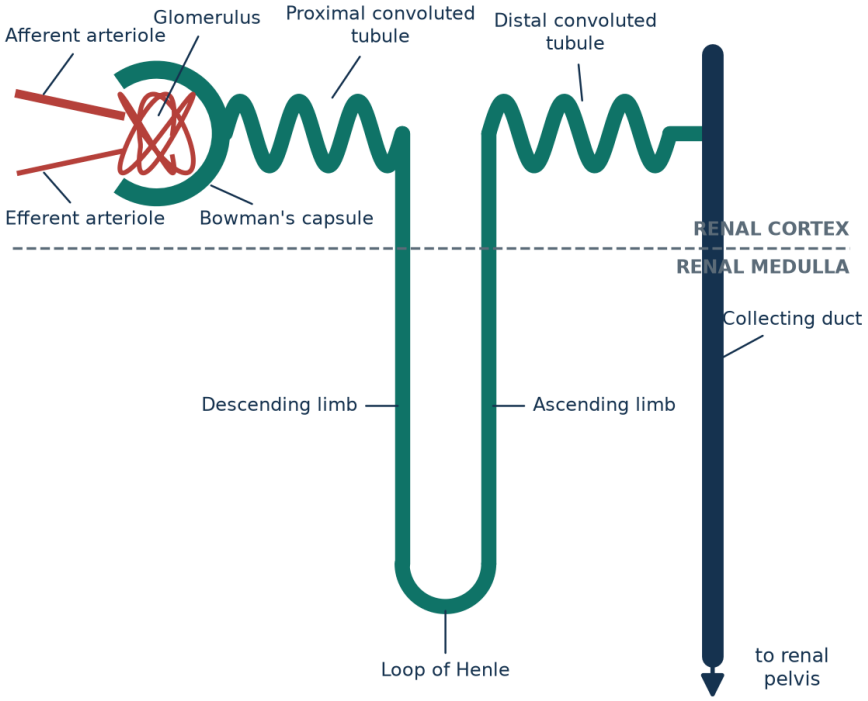
### Collecting duct

The distal convoluted tubules of many nephrons drain into a single collecting duct. As these ducts descend through the medulla, they merge to form larger papillary ducts, which finally discharge into the renal pelvis.

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### Types of nephrons

| Cortical nephrons | Juxtamedullary nephrons |
|---|---|
| They make the majority (about 85%) of nephrons. | They are fewer in number (the remaining, about 15%). |
| They are located primarily within the renal cortex. | They originate deep in the cortex, near the medulla. |
| They have relatively short loops of Henle. | They have long loops of Henle that extend deep into the renal medulla. |




*Structure of a nephron (schematic)*

## 3. Explain the main processes in kidney functioning.

Kidneys filter metabolic wastes and extra water from the blood and excrete them in the form of urine. Urine formation comprises three major steps that take place in the nephron.

### (i) Glomerular filtration

Glomerular filtration (ultrafiltration) is the first step in the formation of urine. It is a physical process. The efferent arteriole is narrower than the afferent arteriole. Due to this difference, the blood within the glomerular capillaries remains under relatively high pressure. This high pressure forces water, urea, glucose, amino acids, vitamins and various salts ($Na^+$, $K^+$, $Cl^-$) from the blood into the Bowman's capsule.

About one-fifth (20%) of the fluid portion of the blood filters into the Bowman's capsule and is called **glomerular filtrate**. Plasma proteins (e.g., albumin and globulins) and blood cells are too large to pass through the capillary walls, so they remain in the blood.

### (ii) Selective reabsorption

The glomerular filtrate contains not only wastes but also vital nutrients that the body cannot afford to lose. Selective reabsorption is the process by which useful materials are transported from the renal tubule back into the peritubular capillaries.

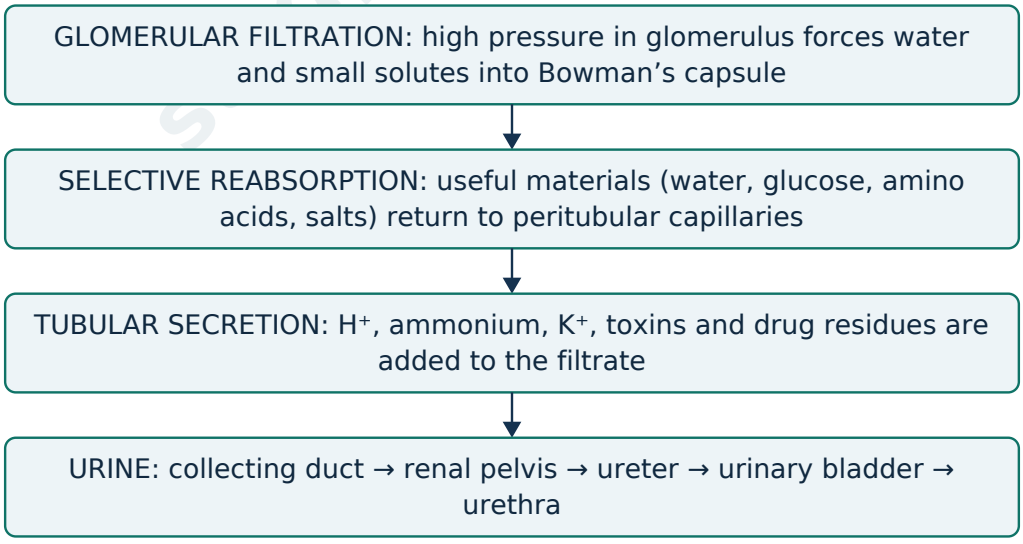
- **Proximal convoluted tubule:** Most reabsorption occurs here. About 75% of the water returns from the filtrate to the capillaries by osmosis. Glucose, amino acids and salts (sodium, potassium and calcium) are returned to the blood by active transport.
- **Descending limb of loop of Henle:** More water is reabsorbed here.
- **Ascending limb:** It is impermeable to water but actively transports salts out of the tubule.
- **Distal convoluted tubule:** Some water and salts are also reabsorbed here.
- **Collecting duct:** It is the final site for water reabsorption.

### (iii) Tubular secretion

Tubular secretion is opposite to reabsorption. In this process, the epithelial cells of the distal convoluted tubule take some materials from the surrounding blood and secrete them into the filtrate. The secreted substances include hydrogen ions, ammonium, potassium ions and various metabolic toxins or drug residues (like penicillin). Through the tubular secretion of hydrogen ions, the kidneys help to maintain the acid–base balance (pH) of the blood.

### Elimination of urine

From the collecting ducts, urine flows through the renal pelvis into a ureter, which leads to the urinary bladder. Muscular contractions of the bladder force the urine out of the body through the urethra. A normal adult eliminates 1 to 2 L of urine a day.




*Steps of urine formation*

## 4. Explain that concentration of urine is regulated by counter-curent and hormonal mechanisms.

### 1. Counter-current mechanism

For the reabsorption of water from the loop of Henle and the collecting duct, nephrons create and maintain a higher concentration of salts (NaCl) in the surrounding fluid. This is done by the counter-current mechanism. It depends on the different properties of the two limbs of the loop of Henle:

- The **descending limb** is permeable to water but not to salts.
- The **ascending limb** is impermeable to water but actively transports salts out of the tubule.

When the filtrate flows through the descending limb, water is reabsorbed into the surrounding fluid because the surrounding fluid has more salts. When the fluid flows through the ascending limb, sodium ions are actively transported from the filtrate to the surrounding fluid, and chloride ions follow the sodium ions. So the salt concentration of the surrounding fluid remains high. As urine passes down the collecting duct through this salty fluid, water moves out by osmosis and enters the blood via the vasa recta. In this way, concentrated urine is formed.

### 2. Hormonal mechanism

When the body needs to conserve water, the kidneys excrete a hypertonic (concentrated) urine. When there is much water in the body, they excrete a hypotonic (dilute) urine. These functions are coordinated by two hormones.

#### (a) Antidiuretic hormone (ADH)

ADH is the body's primary tool for water conservation. It is produced by the hypothalamus and secreted by the posterior pituitary gland.

- **When the body is dehydrated or salty food is taken:** The higher concentration of salts in the blood is detected by osmoreceptors in the hypothalamus. The hypothalamus signals the posterior pituitary to create a sensation of thirst and to release ADH. ADH acts on the walls of the collecting ducts and makes them more permeable to water. More water moves from the filtrate into the hypertonic fluid and then into the blood. So concentrated urine is produced and water is conserved.
- **When the concentration of salts in the blood falls:** The hypothalamus stops ADH secretion. The walls of the collecting ducts become less permeable to water. Less water is reabsorbed and dilute urine is excreted. As a result, the blood volume and blood pressure decrease.

#### (b) Aldosterone

Aldosterone is the hormone responsible for regulating the sodium and potassium levels of urine. It is produced by the adrenal cortex (the outer layer of the adrenal gland present on the top of the kidney). It is secreted when the blood pressure drops or the sodium level in the blood is too low. It acts on the walls of the distal convoluted tubule and the collecting duct and increases the reabsorption of sodium ions from the filtrate into the blood. Due to the higher concentration of sodium ions in the blood, water also moves from the filtrate into the blood.

## 5. Explain the causes and treatments of kidney stones.

Urinary stones are hard, crystalline mineral materials that stick together to form small "pebbles" within the kidney or urinary tract. They may stay in the kidneys or travel out of the body through the urinary tract.

### Causes

- **Calcium oxalate stones (70% of patients):** They are formed due to hypercalciuria (high calcium in urine) and hyperoxaluria (increased level of oxalate in urine).
- **Uric acid stones (10% of patients):** They are caused by hyperuricemia (increased amount of uric acid in blood).
- A high concentration of cysteine and phosphates in urine also causes kidney stones.
- Continuous dehydration of the body increases the chances of kidney stone formation.

### Treatment

Treatment of kidney stones usually involves **lithotripsy**, i.e., the breaking of stones. There are two methods:

- **Extracorporeal Shockwave Lithotripsy (ECSWL):** An instrument called a lithotripter generates shockwaves from outside the patient's body. These shockwaves break the stone into small pieces. Most of the pieces then pass out through urine.
- **Percutaneous Nephrolithotripsy (PNL):** It is used for larger stones. A tube is inserted from the patient's back into the kidney. A small camera called a nephroscope is inserted through the tube to see the stone. Ultrasound equipment is then inserted to break the stone. The stone pieces are grasped with special equipment and pulled out of the kidney.

## 6. Explain in detail the mechanism and problems related to dialysis.

Renal dialysis is the treatment of renal failure in which the metabolic wastes are filtered from the blood by artificial methods. There are two general types of renal dialysis.

### 1. Haemodialysis

In haemodialysis, the wastes are removed by circulating the blood outside the body through an equipment called a **dialyzer**. It is a type of filter and contains a semipermeable membrane made of cellulose. A catheter is inserted into an artery, usually in the arm. The blood is circulated through the dialyzer, which also contains a special fluid called **dialysate**. As the blood passes through the dialyzer, the dialysate attracts the wastes, water and salts out of the blood. They pass through the semipermeable membrane and dissolve in the dialysate. The clean blood then returns to the body through a second catheter inserted in a vein.

### 2. Peritoneal dialysis

Peritoneal dialysis uses the **peritoneal membrane** that lines the abdominal cavity. In this process, 1.5 to 2 litres of dialysate is infused into the patient's abdomen through a catheter. The dialysate is left in the abdomen for 4 to 6 hours. Waste products and excess fluid from the blood capillaries of the peritoneum pass into the dialysate. Then the dialysate is drained out of the abdomen and discarded. This process is repeated several times a day to remove enough wastes and excess fluid.

#### Problems associated with dialysis

Dialysis can be life-saving for people with kidney failure, but it has several problems:

- Dialysis requires access to the bloodstream, which increases the risk of infections.
- Hypotension (low blood pressure) is a common complication of haemodialysis.
- Muscle cramps can occur during dialysis due to changes in salt levels.
- Dialysis can cause anaemia, which leads to fatigue and weakness.
- If too much fluid is removed too quickly, it can lead to dehydration or low blood pressure.
- Dialysis treatments can be emotionally challenging. Patients may feel anxious, depressed or overwhelmed by the demands of the treatment.

## 7. Describe the principles and the problems associated with kidney transplant.

#### Principles

Kidney transplant is a surgical procedure in which a person's non-functioning kidney is replaced with a healthy kidney from a donor. This treatment is offered to individuals with end-stage kidney failure. The process requires careful evaluation, matching and monitoring to ensure the best results for both the recipient and the donor. The donor and the recipient are matched on the basis of blood group, tissue type and immune system compatibility.

#### Potential donors

- **Related donor:** A person who is genetically related to the recipient, such as a parent, sibling or child, can be the best donor of a kidney.
- **Unrelated donor:** The donor is not genetically related to the recipient, such as a spouse, friend or co-worker. This type of donation is less common, but it can be successful if the blood group and tissue type of the donor and recipient match.
- **Deceased donor:** Many people register with hospitals for the donation of their organs after death. Their kidneys are carefully matched to recipients on the basis of blood group, tissue type and immune system compatibility.

#### Problems associated with kidney transplant

- **Organ rejection:** The recipient's immune system may recognize the transplanted kidney as a foreign object and attack it. This can occur even if the patient is taking immunosuppressant medications.
- **Infections:** Patients with a suppressed immune system are at a higher risk of developing infections after a transplant.

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- **Side effects of medicines:** Immunosuppressant medications can cause high blood pressure, weight gain and an increased risk of infections and cancers.

## INQUISITIVE QUESTIONS

### 1. Hypothesize kidney stone by studying the urine test of relevant patients.

Doctors diagnose the nature of a kidney stone by studying the urine test of the patient. From the pH of the urine and the substances present in it, the following hypotheses can be made:

| Finding in urine test | Hypothesis about the stone |
|---|---|
| Urine is acidic | Stone of calcium oxalate |
| Urine is alkaline | Stone of calcium phosphate |
| Urine is persistently acidic | Stone of uric acid |
| High calcium (hypercalciuria) and high oxalate (hyperoxaluria) in urine | Stone of calcium oxalate (found in 70% of patients) |

### 2. Describe the importance of kidney donation for the benefit of kidney failure patients.

Kidney donation can be a life-saving gift for someone with end-stage kidney disease. In kidney failure, the kidneys cannot filter wastes from the blood. Such patients depend on dialysis, which must be repeated again and again and has many problems such as infections, low blood pressure, muscle cramps, anaemia and emotional stress. A transplanted healthy kidney does the natural work of filtration and osmoregulation, so the patient can return to a normal life.

Donating a kidney does not have significant long-term health risks for the donor. Studies have shown that kidney donors generally experience no negative impact on their overall health or lifespan. People can also register with hospitals for the donation of their organs after death. So kidney donation saves the life of the patient without harming the donor.

### 3. Name the important kidney transplant centres in your city.

Students should write the centres of their own city. The textbook mentions the following two centres:

- **Sindh Institute of Urology and Transplantation (SIUT), Karachi:** Pakistan's first kidney transplant centre, established in 1985. It is one of the largest and most successful transplant centres in the world.
- **Pakistan Kidney and Liver Institute & Research Center (PKLI&RC), Lahore:** A tertiary-care hospital established in 2017 for medical care, transplantation and research of kidney and liver diseases.

*Other well-known centres (not given in the textbook): Shifa International Hospital, Islamabad; Armed Forces Institute of Urology (AFIU), Rawalpindi; Sheikh Zayed Hospital, Lahore; Institute of Kidney Diseases, Peshawar.*